

RIO GRANDE, Argentina

WMO: 879340

Lat: 53.7803S

Lon: 67.7586W

Elev: 22

StdP: 101.06

Time Zone: -3.00 (W03)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	-7.4	-5.9	-9.8	1.6	-7.3	-7.8	1.9	-5.5	14.6	4.7	13.5	4.2	2.5	230	0.926

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	9.6	19.8	14.7	18.0	13.0	16.7	12.3	15.1	18.8	13.7	17.1	12.6	15.8	11.2	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.1	9.4	17.2	11.8	8.7	15.6	10.8	8.1	14.2	42.0	18.9	38.7	17.3	35.6	15.8	20.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.9	13.6	DB	-11.3	23.4	2.4	1.7	-13.0	24.6	-14.4	25.6	-15.8	26.6	-17.5	27.8	(4)
(5)			WB	-11.8	17.9	2.8	1.3	-13.8	18.9	-15.4	19.6	-16.9	20.4	-18.9	21.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	5.9	10.9	10.7	8.5	6.0	3.1	0.7	1.0	2.1	4.2	6.2	8.1	10.1	(6)	
	DBStd	4.35	2.21	2.86	2.50	2.48	2.87	3.28	2.52	2.09	2.27	2.19	2.31	1.96	(7)	
	HDD10.0	1619	16	22	61	122	216	278	279	244	175	119	64	23	(8)	
	HDD18.3	4526	232	213	306	371	473	528	537	503	425	376	306	256	(9)	
	CDD10.0	135	42	42	13	1	0	0	0	0	0	2	8	26	(10)	
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH23.3	6	3	3	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	6.8	8.6	7.7	7.1	6.5	5.2	4.8	5.2	5.9	6.1	7.5	8.3	8.6	(14)	
(15) Precipitation	PrecAvg	325	32	23	28	35	33	35	23	17	17	16	27	33	(15)	
	PrecMax	486	69	63	77	70	75	109	62	43	44	38	63	74	(16)	
	PrecMin	213	13	5	7	2	6	12	4	0	2	1	4	8	(17)	
	PrecStd	77	17	15	16	18	19	26	18	12	12	10	17	18	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.1	23.8	19.1	15.0	12.1	9.9	8.1	9.9	13.9	16.1	18.2	19.9	(19)	
		MCWB	15.2	17.2	14.5	11.4	9.6	8.5	6.5	7.2	9.8	11.7	13.1	13.7	(20)	
	2%	DB	18.9	20.2	16.1	13.1	10.0	7.9	6.2	7.9	11.2	14.1	16.2	17.9	(21)	
		MCWB	13.7	15.7	12.3	10.1	8.1	6.7	5.1	5.5	7.7	9.5	11.6	12.3	(22)	
	5%	DB	17.1	18.1	14.9	12.0	8.8	6.2	5.2	6.2	9.9	12.9	14.9	16.2	(23)	
		MCWB	12.6	13.7	11.2	9.3	7.3	5.5	4.3	4.5	6.6	8.6	10.3	11.4	(24)	
	10%	DB	15.9	16.1	13.2	10.8	7.8	5.1	4.8	5.2	8.2	11.1	13.1	15.0	(25)	
		MCWB	11.6	12.4	10.2	8.7	6.7	4.4	3.9	4.0	5.7	7.5	9.3	10.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	15.8	17.9	15.1	11.8	10.1	8.5	6.5	7.2	10.1	11.7	14.0	14.2	(27)	
		MCDB	20.2	22.3	18.6	14.1	11.4	9.4	7.5	8.8	13.4	15.9	18.1	18.6	(28)	
	2%	WB	14.3	16.0	12.9	10.7	8.3	6.6	5.2	5.7	8.2	10.1	11.9	13.0	(29)	
		MCDB	18.1	19.6	15.4	12.7	9.4	7.2	5.8	7.0	10.6	13.3	15.8	16.8	(30)	
	5%	WB	13.1	14.4	11.5	9.9	7.4	5.5	4.8	4.8	6.8	8.9	10.6	11.8	(31)	
		MCDB	16.4	17.7	14.0	11.3	8.2	6.1	5.0	5.8	9.0	12.1	14.0	15.5	(32)	
	10%	WB	11.9	12.8	10.7	8.9	6.4	4.4	3.7	4.1	5.9	7.9	9.8	11.0	(33)	
		MCDB	15.3	15.7	12.7	10.4	7.3	5.0	4.3	5.1	7.7	10.6	12.9	14.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.6	9.6	9.1	8.5	7.4	5.8	5.6	6.4	7.9	9.2	9.3	9.6	(35)	
		MCDBR	10.8	12.0	10.4	9.5	8.2	6.0	5.7	7.5	9.6	11.1	11.1	11.4	(36)	
	5% WB	MCWBR	7.2	8.0	7.5	7.3	6.7	5.3	4.9	5.9	6.8	7.4	7.4	7.4	(37)	
		MCWBR	10.4	11.4	9.6	8.9	7.3	5.8	5.2	6.9	8.8	10.4	10.7	10.9	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.319	0.313	0.312	0.302	0.285	0.270	0.281	0.299	0.309	0.314	0.319	0.322	(40)	
		taud	2.553	2.575	2.562	2.550	2.457	2.364	2.420	2.491	2.520	2.516	2.514	2.523	(41)	
	Ebn at Noon		984	953	890	790	669	597	659	776	877	941	978	990	(42)	
		Edh at Noon	99	90	79	62	48	42	48	64	80	94	102	104	(43)	
(44) All-Sky Solar Radiation	RadAvg		5.73	4.76	3.15	1.80	0.92	0.58	0.72	1.44	2.73	4.35	5.58	6.14	(44)	
	RadStd		0.28	0.23	0.23	0.10	0.07	0.04	0.05	0.10	0.18	0.17	0.28	0.36	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page